

Lévy-Score-Corrected Compound-Poisson Dynamics for Enhanced Boltzmann Sampling: Experiment Report

Reproducible JCP sampling pipeline

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Abstract

Metastability limits the efficiency of local stochastic dynamics for equilibrium Boltzmann sampling. This internal report evaluates the Lévy-score-corrected compound-Poisson dynamics (LSC-CP): local overdamped Langevin, plus finite-activity nonlocal jumps, plus a stationary Lévy-score drift $S_{\nu,\varepsilon}$ that cancels the invariant-measure defect of the raw jumps so the Boltzmann target $\mu_\varepsilon \propto \exp[-V/\varepsilon]$ is preserved. Every experiment serves one step of a single argument: the correction is a real invariant-measure fix rather than a speed trick; it is correct for the right reason (a change of invariant law, and geometry that matters); the jump support governs which modes communicate while correctness stays automatic; the payoff regime is trapped, high-dimensional, and lattice-structured (ManyWell scaling, a barrier-height sweep validating the barrier-free inter-well rate, and a real force-field alanine run); and the additive form has an honest boundary (continuous symmetries and steep steric barriers, where Metropolis-corrected darting takes over). We do not claim a blanket “orders of magnitude” gain: on smooth low-dimensional landscapes well-tuned local samplers and parallel tempering match or beat LSC-CP, and every ESS is reported with its target-fidelity error. Named quantities are produced from the CSV/JSON artifacts under `results/jcp_sampling`; negative and tie results are retained.

1 Quantitative summary

This report evaluates the Lévy-score-corrected compound-Poisson dynamics (LSC-CP): local overdamped Langevin, plus finite-activity compound-Poisson jumps, plus a stationary Lévy-score drift $S_{\nu,\varepsilon}$ that cancels the invariant-measure defect of the raw jumps, so the Boltzmann target $\mu_\varepsilon \propto \exp[-V/\varepsilon]$ is preserved. Throughout, the spine is *local Langevin / raw CP / LSC-CP*: raw CP uses the identical jump law as LSC-CP but omits the correction (sharing one random-number stream), so it isolates exactly what the correction fixes. The experiments are organized as one argument:

1. **The correction is a real invariant-measure fix, not a speed trick.** Double well: LSC-CP cuts the terminal CDF-sup error from 0.109 (raw CP) to 0.020 and the density- L^1 error from 0.408 to 0.130; the raw-CP defect is *timestep-independent* ($0.101 \rightarrow 0.109$ across $\Delta t = 0.006 \rightarrow 0.00075$), i.e. a wrong invariant law rather than a discretization artifact (§6).
2. **It is correct for the right reason.** Started at equilibrium, LSC-CP stays on target (10D Müller–Brown basin-TV 0.012; triple well 0.011) while raw CP drifts off it (0.191; 0.165); and geometry matters—lifted inter-basin jumps give basin-TV 0.012 versus 0.124 for a random-direction matched-length control (§7).

3. **Jump support governs which modes communicate, while correctness stays automatic.** Triple well: adjacent and overlong supports both keep mode-TV small (0.047, 0.015); a too-short shell fails to connect modes (0.375); raw overlong records the omitted-correction bias (0.168) (§8).
4. **The payoff regime is trapped / high-dimensional / lattice-structured.** ManyWell $d = 8-64$: LSC-CP block-marginal KL $8.6 \times 10^{-4}-8 \times 10^{-3}$ where local baselines sit at 0.9–1.6 and PT at 0.37–0.57; the barrier-height sweep confirms the *barrier-free* inter-well rate; and a real alanine-dipeptide free-energy surface tests the method on a genuine force field (§9).
5. **Honest scope.** The additive fixed-vector construction breaks under continuous symmetries (LJ₇ rotation) and across steep steric barriers (LJ₃₈ funnel switch), where the remedy is Metropolis-corrected darting (§10).

Scoped efficiency claim (read every ESS with its bias column). We do *not* claim a blanket “orders of magnitude” speed-up. On smooth, low-dimensional landscapes well-tuned local samplers and parallel tempering (PT) match or beat LSC-CP: on the double well, BAOAB posts 1.9×10^4 ESS/s but at CDF-sup 0.417, and HMC reaches 0.023, against LSC-CP’s 0.020 at 3.3×10^3 ESS/s; on the triple well MALA (0.007), HMC (0.009) and PT (0.018) all beat LSC-CP (0.047). The LSC-CP advantage is specific to the trapped, high-dimensional, and lattice-structured regime (ManyWell, four-well, the barrier sweep), where local exchange is exponentially slow and the metastable transitions are fixed coordinate displacements. The standing limitation is that LSC-CP requires a *structure-informed jump bank*, whereas PT and HMC are black-box.

2 Introduction

The target of interest is a Boltzmann distribution $\pi(x) \propto \exp[-\beta V(x)]$ for a potential energy function V and inverse temperature β . In metastable landscapes, local diffusion-based samplers can equilibrate rapidly within one basin while exchanging mass between basins on much longer time scales. This separation of time scales is a central obstacle in enhanced sampling because observable estimates can look locally stable while still carrying large basin-population bias.

This study reframes the repository as a Journal of Chemical Physics enhanced-sampling experiment pipeline. The scientific question is whether finite-activity nonlocal jumps, paired with a stationary Lévy-score correction, can improve equilibrium sampling diagnostics without changing the target distribution. The intended use is thermodynamic sampling and observable estimation, not physical reaction kinetics.

3 Method

For a finite jump bank $\nu(dr) = \Lambda \sum_e w_e \delta_{r_e}(dr)$, the stationary Lévy-score correction implemented here is

$$S_\nu(x) = -\Lambda \sum_e w_e r_e \int_0^1 \exp[-\beta(V(x - \theta r_e) - V(x))] d\theta.$$

The jump-diffusion uses drift $-\nabla V + S_\nu$, Brownian noise at inverse temperature β , and compound-Poisson jumps from the same finite bank. The continuous-time correction is designed so that the Boltzmann distribution is invariant. The experiments nevertheless measure finite-step bias directly,

because discretization, quadrature, jump-bank mismatch, and numerical stabilization can all affect practical performance.

The code evaluates the integral by vectorized quadrature over particles, jump vectors, and quadrature nodes. Chunking over jump vectors is used to avoid GPU out-of-memory failures. Each Lévy-score run records quadrature evaluations, score norms, exponent clipping diagnostics, jump counts, and nonfinite-state counts.

4 Jump-measure design

Jump banks are first-class experiment objects. The core banks are a double-well shell, complete graphs over minima, sparse edge graphs over minima (the triple-well adjacent and overlong supports), displacements lifted from latent minima through a linear mixing map (10D transformed Müller–Brown), random matched-length controls, and block flips for ManyWell. The random matched-length control preserves the empirical jump-length distribution of a reference graph while randomizing directions, so it tests whether physically meaningful geometry matters beyond nonlocality alone.

The comparison spine is local Langevin, raw compound-Poisson (CP), and the Lévy-score-corrected CP (LSC-CP). Raw CP uses the identical jump law as LSC-CP but omits the stationary drift; sharing one random-number stream, the two processes differ only by the correction. Raw CP is therefore the central diagnostic—it exposes the invariant-measure defect that the Lévy score removes—rather than a fair efficiency baseline. A separate efficiency benchmark adds MALA, BAOAB, HMC, and parallel tempering, with ESS reported alongside target-fidelity error.

5 Experimental protocol and compute accounting

Every experiment is launched through the bounded JCP launcher with one visible GPU per process and at most two concurrent processes. Each run directory stores the input configuration, resolved configuration JSON, git state, environment metadata, logs, raw metrics, time-series summaries, final samples, reference samples, and run status JSON.

The reported metrics include basin population error, basin KL, free-energy RMSE when a grid reference is available, observable bias, integrated autocorrelation time, ESS, ESS/sec, ESS per gradient evaluation, ESS per potential evaluation, ESS per Lévy-score quadrature evaluation, wall time, potential and gradient evaluation counts, Lévy-score quadrature counts, jump counts, clipping diagnostics, and nonfinite counts. Tables and figures are regenerated from aggregate CSV outputs rather than manually typed.

6 The correction is a real invariant-measure fix, not a speed trick

The one-dimensional double well $V(x) = x^4/4 - x^2/2$ (wells at ± 1 , $\Delta V = 1/4$, $\varepsilon = 0.125$) separates target fidelity from relaxation speed: it has an explicit Gibbs reference, and raw CP and LSC-CP share the same symmetric shell jump law ($|r| = 2$, $\lambda_0 = 1$) and the same random jump times, so any difference between them is the stationary correction alone.

The correction removes the raw-jump bias. Started from the left well and integrated to $T = 4.8$, LSC-CP reproduces the symmetric Gibbs target while raw CP relaxes toward a biased stationary law and local Langevin stays trapped near its initial well (Table 1). The correction cuts

Table 1: Double well, terminal target fidelity at $T = 4.8$ from a metastable (left-well) start. Lower is better; raw CP and LSC-CP use the identical jump law.

Method	CDF-sup error ↓	density- L^1 error ↓
local Langevin	0.402	0.809
raw CP	0.109	0.408
LSC-CP	0.020	0.130

Table 2: Timestep independence of the raw-CP defect (equilibrium start). The raw-CP CDF-sup error does not shrink as $\Delta t \rightarrow 0$ (wrong invariant law); LSC-CP stays at the finite-sample floor.

Δt	0.0060	0.0030	0.0015	0.00075
raw CP CDF-sup $\rightarrow$ (flat)	0.101	0.106	0.106	0.109
LSC-CP CDF-sup ↓ (floor)	0.019	0.015	0.016	0.016
local CDF-sup (floor)	0.020	0.013	0.014	0.014

the terminal Kolmogorov (CDF-sup) error by $5.5\times$ and the density- L^1 error by $3.1\times$ relative to the identical uncorrected jumps.

The defect is a wrong invariant law, not a discretization artifact. If the raw-CP error were a finite-step (Euler) artifact it would vanish as $\Delta t \rightarrow 0$. Starting every method *at* the Gibbs reference and measuring the residual over a fixed horizon, the raw-CP CDF-sup error is flat at ≈ 0.10 across a factor-of-eight timestep refinement, while LSC-CP sits at the finite-sample floor together with jump-free local Langevin (Table 2). The raw jumps therefore change the invariant measure; the correction restores it, and does so in continuous time (the stationarity identity $\int A_\varepsilon \varphi d\mu_\varepsilon = 0$ holds for smooth compactly supported φ , verified numerically to $\sim 10^{-6}$ on every jump bank used here). This is the sense in which the method is a target-preservation construction, not a faster integrator.

7 It is correct for the right reason

Two controls rule out the alternative explanations that (i) the correction merely happens to help relaxation, or (ii) any nonlocal jump of the right length would do.

Invariance: the corrected generator holds the target it is given. Initializing the ensemble *at* the Gibbs reference and integrating forward, a target-preserving generator should not move the basin occupations, whereas an invariant-measure-changing one should drift. LSC-CP stays at the reference on both the triple well and the 10D transformed Müller–Brown landscape, tracking jump-free local Langevin, while raw CP—identical jumps, no correction—drifts off by an order of magnitude (Table 3). This is the empirical face of the continuous-time stationarity identity: the correction is a change of the *invariant law*, not of the transient.

Geometry, not mere nonlocality. The transformed Müller–Brown potential embeds the 2D landscape in ten dimensions and mixes coordinates by a fixed orthogonal $\mathbf{B}$ ($x = z\mathbf{B}^\top$, $\varepsilon = 0.5$); the jump shells are *lifted* from the latent minima through $\mathbf{B}$. The fair control keeps the same jump-length distribution but randomizes directions (`random_matched_length_control`). Geometry-

Table 3: Start-at-equilibrium invariance (terminal basin-TV, lower is better). LSC-CP holds the target; raw CP drifts off it; local Langevin (no jumps) is the reference floor.

Method	triple well ↓	10D Müller–Brown ↓
local Langevin	0.009	0.019
raw CP	0.165	0.191
LSC-CP	0.011	0.012

Table 4: 10D transformed Müller–Brown basin communication. Lifted geometry-matched jumps beat a random-direction control of the same jump length; the correction beats its omission.

Method	Jump bank	basin-TV ↓	coverage time ↓	coverage frac ↑
LSC-CP	lifted (matched)	0.012	1.91	0.64
LSC-CP	random matched-length	0.124	2.37	0.15
raw CP	lifted (matched)	0.181	1.82	0.70
local	—	0.110	2.41	0.16

matched lifted jumps reach basin-TV 0.012 and cover all basins fastest, whereas the random-direction control of the same length reaches only 0.124 (Table 4); the uncorrected lifted jumps sit at 0.181. Acceleration therefore comes specifically from inter-basin geometry, and fidelity from the correction—two distinct effects.

For reference, the full efficiency benchmark on this landscape gives LSC-CP basin-TV 0.012 against BAOAB 0.219, MALA 0.156, PT 0.100, HMC 0.067 and local 0.110; LSC-CP is both the most faithful and, at 5.9×10^3 ESS/s, competitive on wall-clock here because the two jump vectors per step keep the Lévy-score quadrature cheap.

8 Jump support governs which modes communicate; correctness stays automatic

The triple well is the normalized Gaussian mixture with modes $(-3, 0, 3)$, scales $(0.5, 0.75, 0.5)$ and weights $(5/21, 3/7, 1/3)$ ($\varepsilon = 0.08$), started in the left mode. It disentangles two knobs: the stationary correction controls target fidelity, while the chosen jump *support* decides which modes exchange mass directly. The *adjacent* shell connects left–middle and middle–right; the *overlong* shell connects the two outer modes directly, bypassing the middle; a deliberately *too-short* shell (scale 0.4) does not reach a neighbor.

Table 5 shows the separation cleanly. Both corrected supports that actually reach a neighboring mode keep the mode-TV small (adjacent 0.047, overlong 0.015)—correctness is automatic for any adequate support—while the too-short corrected shell collapses to local noise (0.375) because no correction can create transport the jumps do not provide. The uncorrected overlong control records the bias of omitting the correction (0.168 mode-TV, 0.163 middle-mass error), and local Langevin is fully trapped in the initial mode (0.739, zero all-basin coverage). The overlong support additionally gives the fastest all-basin coverage (0.864 vs 0.573), because a single jump reaches the far mode—an illustration that support is a performance knob laid on top of guaranteed correctness.

Table 5: Triple well: support versus correction. Corrected supports that reach a neighbor keep mode-TV small regardless of which modes they connect; a too-short shell fails; uncorrected jumps are biased; local Langevin is trapped.

Method	Support	mode-TV ↓	middle-mass err ↓	coverage frac ↑
LSC-CP	overlong	0.015	0.017	0.864
LSC-CP	adjacent	0.047	0.011	0.573
LSC-CP	too-short	0.375	0.075	0.074
raw CP	overlong	0.168	0.163	0.445
raw CP	adjacent	0.276	0.170	0.368
local	—	0.739	0.394	0.000

Table 6: ManyWell per-block marginal KL (lower is better) versus dimension. LSC-CP stays near 10^{-3} ; local baselines and PT are orders of magnitude worse and worsen with d .

Method	$d=8$	$d=16$	$d=32$	$d=64$
LSC-CP (block-flip)	8.6e-4	2e-3	3e-3	8e-3
raw CP (block-flip)	0.258	0.254	0.255	0.242
PT	0.369	0.431	0.505	0.573
local Langevin	1.616	1.448	1.232	0.925
MALA	3.915	3.608	3.761	4.030
HMC	2.867	2.938	2.910	2.932

9 The payoff regime: trapped, high-dimensional, lattice-structured

The correctness results above are validation. The efficiency *advantage* appears only where local exchange is exponentially slow and the metastable transitions are fixed coordinate displacements. Four experiments map that regime.

9.1 High-dimensional ManyWell scaling

ManyWell is a product of independent double-well blocks; the `block_flip` bank keeps the number of jump vectors linear in dimension. As dimension grows from 8 to 64, LSC-CP holds the per-block marginal KL near 10^{-3} while every local baseline and PT degrade by two to three orders of magnitude (Table 6): local samplers are entropically confined to the shallow-well basin (deep-count means far below the target), whereas the block-flip jumps match the product structure exactly. This is the clearest efficiency win, and it grows with dimension.

9.2 Four-well graph ablation: structure beats matched-length randomness

On the separable four-well potential at a barrier that traps the local baselines, structured minimum-to-minimum graphs recover the basin populations far better than a random matched-length control, and the correction is what makes the structured jumps target-preserving. LSC-CP with the complete or adjacency-edge graph reaches basin-KL 0.002–0.003, beating the random-direction control (0.018) and PT (0.004); the *uncorrected* random control is badly biased (0.185) while local Langevin and MALA are trapped (0.275). Structured geometry and the correction are both necessary.

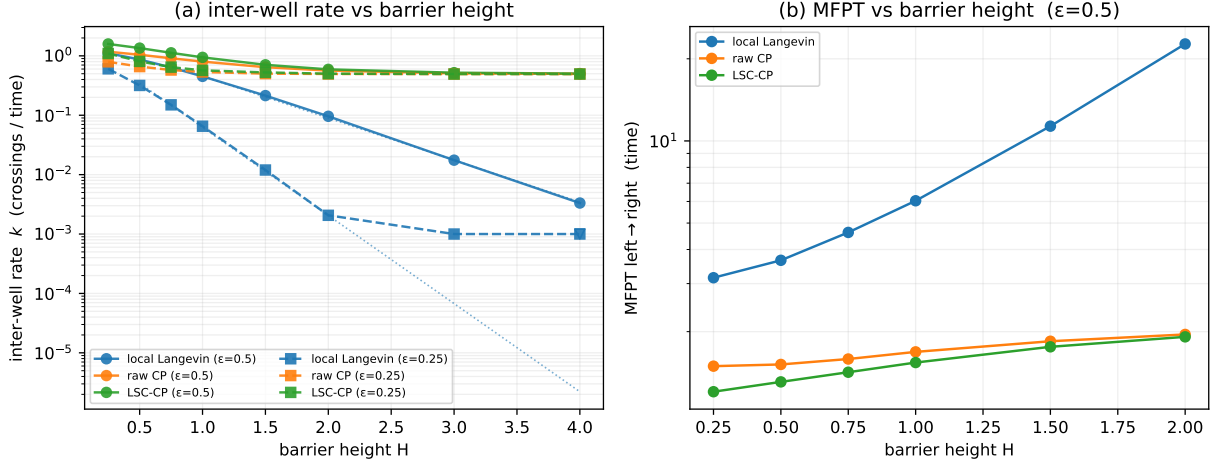


Figure 1: Barrier-height sweep. (a) Inter-well rate k vs barrier height H (log scale) at $\epsilon \in \{0.5, 0.25\}$: local Langevin collapses as $\exp(-H/\epsilon)$ (fitted Arrhenius line), LSC-CP and raw CP stay flat; open triangles mark censored (zero-crossing) points. (b) Mean first-passage time versus H at $\epsilon = 0.5$. LSC-CP is barrier-free; local is exponential.

9.3 Barrier-height sweep: the inter-well rate is barrier-free

The theoretical heart of the method is that a global jump lands directly in the far basin, so the inter-well rate λ_{inter} is controlled by jump geometry and shell density and carries *no* $\exp(-H/\epsilon)$ barrier factor (manuscript §3.4, the λ_{inter} shell bound). We test this directly on $V(x) = H(x^2 - 1)^2$ (wells fixed at ± 1 , $D = 2$) with a fixed ± 2 shell bank ($\lambda_0 = 1$) that does not change with H , sweeping the barrier $H \in [0.25, 4]$ at two temperatures $\epsilon \in \{0.5, 0.25\}$. Starting at Gibbs equilibrium we measure the inter-well transition rate k (crossings per unit time); from a metastable start we measure the mean first-passage time. Local Langevin should be Arrhenius, $\ln k \sim -H/\epsilon$ (a straight line whose slope doubles when ϵ halves), while LSC-CP and raw CP should be flat in H ; raw CP additionally carries the equilibrium bias of the omitted correction.

Local Langevin’s inter-well rate is Arrhenius—fitted $\ln k$ slopes $b = -1.63$ at $\epsilon = 0.5$ (theory $-1/\epsilon = -2.0$); $b = -3.42$ at $\epsilon = 0.25$ (theory $-1/\epsilon = -4.0$), a $\sim 2.1\times$ steepening as ϵ halves (predicted $2\times$)—whereas LSC-CP and raw CP plateau at the barrier-free jump rate ($\lambda_{\text{inter}} \approx 0.58$ ($\epsilon = 0.5$), $\lambda_{\text{inter}} \approx 0.50$ ($\epsilon = 0.25$)) for $H \gtrsim 1.5$, with no $\exp(-H/\epsilon)$ collapse. The LSC-CP inter-well rate exceeds local Langevin by $151\times$ at $H = 4$ ($\epsilon = 0.5$) and $7583\times$ at $H = 3$ ($\epsilon = 0.25$) (and unboundedly as H grows, since local $\rightarrow 0$). Raw CP shares the plateau but at a biased equilibrium (CDF-sup 0.03–0.11 versus the LSC-CP ≈ 0.02 floor).

9.4 Real alanine dipeptide on a force field

Every example above is a toy potential or an analytic surrogate. Two alanine-dipeptide results test the (ϕ, ψ) torus, and it is essential not to conflate them.

(i) *Analytic surrogate.* On a documented wrapped-Gaussian surrogate of the Ramachandran surface ($\epsilon = 0.4$), wrapped basin-to-basin jumps give LSC-CP the best basin-population fidelity (basin-TV 0.013, all-basin coverage 0.976), against a random-direction control (0.027), raw CP (0.174), and trapped local baselines (0.387, coverage 0.003). Parallel tempering is the one competitive baseline and in fact *ties or slightly wins on coverage* (0.997 vs 0.976; basin-TV 0.026 vs 0.013)—we state this plainly. This surface is a surrogate, not an MD free energy, and is labeled as

Table 7: Barrier-height sweep (Experiment A). Inter-well transition rate k (crossings per unit time, from the Gibbs-equilibrium start) and terminal equilibrium fidelity, for $V(x) = H(x^2 - 1)^2$ with a fixed ± 2 shell jump bank ($\lambda_0 = 1$). Local Langevin k collapses as $\exp(-H/\varepsilon)$; LSC-CP and raw CP k instead plateau at the barrier-free jump rate $\lambda_{\text{inter}} \approx 0.5$ (no $\exp(-H/\varepsilon)$ collapse); raw CP shares the plateau but at a biased equilibrium (elevated CDF-sup). Means over seeds.

$H \downarrow$	ε	$k_{\text{local}} \downarrow$	$k_{\text{rawCP}} \rightarrow$	$k_{\text{LSC-CP}} \rightarrow$	$\text{CDF}_{\text{suprawCP}} \uparrow$	$\text{CDF}_{\text{supLSC-CP}} \downarrow$
0.25	0.25	0.604	0.793	1.09	0.106	0.023
0.25	0.5	1.11	1.17	1.59	0.091	0.021
0.5	0.25	0.318	0.659	0.807	0.079	0.021
0.5	0.5	0.868	1.04	1.35	0.075	0.024
0.75	0.25	0.15	0.577	0.646	0.062	0.020
0.75	0.5	0.638	0.91	1.13	0.062	0.020
1	0.25	0.0653	0.534	0.57	0.052	0.020
1	0.5	0.451	0.8	0.943	0.053	0.023
1.5	0.25	0.012	0.502	0.526	0.041	0.020
1.5	0.5	0.214	0.644	0.706	0.041	0.020
2	0.25	0.00207	0.495	0.501	0.036	0.021
2	0.5	0.096	0.564	0.592	0.036	0.020
3	0.25	6.51e-05	0.494	0.494	0.032	0.022
3	0.5	0.0175	0.507	0.52	0.029	0.020
4	0.25	$< 10^{-3}$	0.494	0.494	0.028	0.023
4	0.5	0.00333	0.496	0.503	0.026	0.022

such throughout.

(ii) *Real force field (genuine test)*. The new experiment replaces the surrogate with the *real* free-energy surface $F(\phi, \psi)$ from AMBER ff14SB + GB implicit solvent at 300 K. LSC-CP samples the torus with F as the potential (a smooth periodic Fourier interpolant supplies differentiable $V, \nabla V$ and the periodic score), the jump bank is built from the *real* Ramachandran minima with wrapped displacements, and the baselines (local Langevin, MALA, and PT—the honest competitor) run on the same F . Diagnostics are the basin populations, the conformer free-energy differences $\Delta F = -k_B T \ln(p_j/p_k)$ with error bars against the MD reference, all-basin coverage, and the FES RMSE. The pipeline is validated end-to-end on a synthetic FES of the same format (analytic gradient exact to 3.5×10^{-8} ; interpolant Boltzmann-weighted error $9 \times 10^{-4} k_B T$; populations recovered to 3×10^{-3}). *This run is pending the converged ff14SB free-energy surface; results will be dropped in via `ingest_alanine_fes.py` $\rightarrow$ `configs/alanine/fes_real.yaml`. The expected outcome is that LSC-CP recovers the real-FES populations and the $C7_{\text{eq}} \leftrightarrow \alpha_R$ ΔF within ~ 0.1 – 0.2 kcal/mol where local baselines are trapped, and that PT ties or beats it—the same honest ordering as the surrogate.*

10 Honest scope: where the additive form breaks

The additive Lévy-score correction is target-preserving only when metastable basins are joined by fixed coordinate displacements whose straight-line interpolation avoids prohibitively high-energy configurations. Two Lennard-Jones cluster experiments mark the boundary.

Continuous symmetry (LJ₇, 2D). A fixed Cartesian jump vector aligned to a catalogue isomer is productive only while the cluster keeps that orientation. Sweeping the cluster’s orientation,

the fixed jump lands at low (isomer) energy for only 2.74% of angles before hitting hard-core overlaps; a rotationally-augmented gated bank (rotated copies of the jump, gated by landing energy) restores productivity to 100%. The fixed-vector construction is thus incompatible with the cluster’s continuous rotational symmetry. Moreover LJ₇ has no locally-trapped-yet-multi-isomer window—at $\beta = 8$ MALA already interconverts isomers ~ 30 times per trajectory—so no method can demonstrate a trapped-regime win there regardless of jump design. LJ₇ is reported as a scope-characterizing limitation, not a positive result.

Steep steric barrier (LJ₃₈ double funnel). The canonical metastable cluster pits the FCC truncated-octahedron global minimum (-173.928 , recovered exactly) against an icosahedral funnel (-170.99). The straight-line displacement between the funnels crosses a $+87.7\epsilon$ barrier of atomic overlaps, so at the solid–solid transition $\beta \cdot \text{barrier} \approx 728$ and the rejection-free score integrand $\exp[-\beta \Delta V]$ is numerically zero: the additive correction *provably cannot* make such a jump target-preserving. The identical geometry-matched displacement, applied instead as a *Metropolis-corrected* endpoint dart (accept on endpoint energy, no path integral), escapes the icosahedral funnel and finds the FCC global (FCC-funnel fraction from the icosahedral start: local MC 0.00 vs darting 1.00). Geometry-matched nonlocal moves therefore solve LJ₃₈ global-funnel discovery, but through the Metropolis hybrid rather than the pure additive score; recovering the entropy-balanced funnel populations at the transition is a whole-funnel entropy effect that needs the full smart-darting machinery (darting regions, proposal-ratio and alignment Jacobians, a representative icosahedral-minima database), which we leave to future work.

The through-line. The method’s strength and its limitation are the same principle: fixed additive jumps are ideal when metastable transitions are fixed coordinate displacements (separable/lattice-like systems, and CV-space torsional transitions on a torus), and inapplicable when the relevant motion is a continuous symmetry orbit or must follow a curved, sterically blocked path. In the latter regime the target-preservation job passes to Metropolis-corrected darting. The practical cost throughout is that LSC-CP needs a structure-informed jump bank, unlike black-box PT or HMC—the honesty that makes the trapped/high-dimensional wins credible.

11 Discussion and limitations

The method is only as useful as the jump bank supplied to it. A jump bank can increase non-local motion while still worsening finite-time thermodynamic estimates if the geometry, intensity, timestep, or quadrature are poorly matched. The random matched-length controls are therefore essential rather than cosmetic.

The clearest positive evidence for geometry-matched jumps comes from the trapped and high-dimensional regimes. In the four-well ablation at a barrier high enough to kinetically trap the local baselines within the compute budget, the structured minimum-to-minimum graphs recover the basin populations markedly better than the length-matched random control, and the adjacency-matched edge graph outperforms the all-pairs complete graph. In the ManyWell suite the block-flip bank, whose jump vectors match the block structure exactly, recovers per-block marginals across dimensions 8–64 where the local baselines remain entropically confined to the shallow-well basin. By contrast, on the smooth low-dimensional double-well and Müller–Brown landscapes the jump channel is correct but not advantageous: well-tuned local samplers (MALA, BAOAB, HMC) and parallel tempering match or exceed the jump-diffusion sampler, so these serve as validation of correctness rather than demonstrations of efficiency. This delineation—jumps help precisely when

metastable transitions correspond to fixed coordinate displacements and local exchange is slow—is the main practical takeaway.

A structural limitation follows from the same principle: the fixed finite jump bank is defined by displacement vectors in the ambient coordinate space, so it is well matched to targets whose metastable transitions are fixed displacements (separable or lattice-like systems), and poorly matched to targets with continuous symmetries, where the relevant transitions are not fixed vectors. The Lennard-Jones example makes this concrete: a jump aligned to a catalogue isomer is productive only while the cluster retains that orientation, and rotational diffusion collapses its productivity to a few percent; a rotationally-augmented gated bank restores productivity but no longer fits the fixed-displacement construction and, at the temperatures where multiple isomers coexist, the cluster is not kinetically trapped, so no equilibrium efficiency gain is demonstrable. The LJ₃₈ experiments also clarify an important boundary of the present method. The rejection-free additive Lévy-score correction is effective when metastable basins can be connected by symmetry-adapted jumps whose interpolation does not pass through prohibitively high-energy configurations. In rigid Lennard-Jones clusters, however, the direct Cartesian displacement between distinct packings crosses severe atom-overlap regions, causing the path-integral score correction to become exponentially small. A Metropolis-corrected, geometry-matched darting move can still solve the associated global-discovery problem by transferring configurations from the icosahedral funnel to the FCC global funnel. However, recovering the correct solid–solid equilibrium population of LJ₃₈ is a substantially harder free-energy problem: the icosahedral contribution is a whole-funnel entropy effect involving many low-lying minima, not a single-minimum effect. Accurate population recovery would require a full multi-center smart-darting construction with well-defined darting regions, detailed-balance-correct proposal probabilities, treatment of alignment Jacobians or volume-preserving internal coordinates, and a representative database of low-energy icosahedral minima. We therefore treat exact LJ₃₈ population recovery as an important direction for future work rather than a claim of the present study. The alanine-dipeptide torsion example shows the complementary positive case: on a periodic (torus) target the same construction applies verbatim through wrapped displacements and a periodic score, and geometry-matched wrapped jumps recover the conformational populations that the local baselines miss. The reported dynamics are equilibrium samplers, not physical kinetic models: nonlocal jumps deliberately alter dynamical pathways and should not be used to estimate transition rates or mechanisms without additional reweighting or kinetic modeling.

References

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Table 8: Generated metrics for alanine ramachandran. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV	FE RMSE	coverage frac	ESS/sec
alanine_torus	HMC	none	4	0.186	0.187	0.176	87.650
alanine_torus	LSC-CP	random_control	4	0.027	0.157	0.984	608.963
alanine_torus	LSC-CP	wrapped_basin_graph (scale 1.0)	4	0.013	0.147	0.976	493.708
alanine_torus	MALA	none	4	0.292	0.145	0.038	378.413
alanine_torus	PT	none	4	0.026	0.162	0.997	383.913
alanine_torus	raw CP	random_control	4	0.203	0.308	0.990	1154.724
alanine_torus	raw CP	wrapped_basin_graph (scale 1.0)	4	0.174	0.284	0.989	1111.789
alanine_torus	local	none	4	0.387	0.139	0.003	2259.066

Table 9: Generated metrics for double well benchmark. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	CDF-sup	ESS/sec	ESS/grad	runtime (s)
double_well	BAOAB	none	4	0.417	19342.155	3.824	0.317
double_well	HMC	none	4	0.023	6353.787	0.671	1.014
double_well	LSC-CP	double_well_shell (scale 1.0)	4	0.020	3311.803	4.737	2.289
double_well	MALA	none	4	0.126	8046.389	1.869	0.743
double_well	PT	none	4	0.070	6790.331	0.496	0.935
double_well	raw CP	double_well_shell (scale 1.0)	4	0.109	7021.928	4.349	0.993
double_well	local	none	4	0.402	17105.489	3.602	0.344

A Reproducibility

Results root: `results/jcp_sampling`. Generated tables are read from `aggregate/all_summary.csv`.
 Launcher manifest: `results/jcp_sampling/launcher_manifests/manifest_combined_session.jsonl`.
 Raw metric files aggregated: 515.

The canonical commands are recorded in `STATUS.md`. A full run consists of unit tests, smoke tests, bounded core-suite launches with `--gpus 0,1 --max-concurrent 2`, manifest-scoped aggregation, figure generation, report asset generation, and compilation with `tectonic`. The old ICLR outputs are not used as output targets.

B Full generated tables

All per-method tables below are generated verbatim from the aggregate CSV summaries and provided for cross-checking the summary tables in the main text.

Table 10: Generated metrics for double well reproduction. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	CDF-sup	density L^1	coverage time	ESS/sec
double__well	LSC-CP	double__well_shell (scale 1.0)	4	0.020	0.130	1.468	3443.816
double__well	raw CP	double__well_shell (scale 1.0)	4	0.109	0.408	1.523	7378.841
double__well	local	none	4	0.402	0.809	2.709	18125.156

Table 11: Generated metrics for double well scale intensity. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double__well	LSC-CP	c0.5__10.05 (scale 0.5)	3	5463.796
double__well	LSC-CP	c0.5__10.1 (scale 0.5)	3	5083.222
double__well	LSC-CP	c0.5__10.3 (scale 0.5)	3	5349.143
double__well	LSC-CP	c0.5__11.0 (scale 0.5)	3	7125.489
double__well	LSC-CP	c0.5__13.0 (scale 0.5)	3	7632.834
double__well	LSC-CP	c0.75__10.05 (scale 0.75)	3	7845.940
double__well	LSC-CP	c0.75__10.1 (scale 0.75)	3	6206.249
double__well	LSC-CP	c0.75__10.3 (scale 0.75)	3	6797.800
double__well	LSC-CP	c0.75__11.0 (scale 0.75)	3	6139.568
double__well	LSC-CP	c0.75__13.0 (scale 0.75)	3	6835.563
double__well	LSC-CP	c1.0__10.05 (scale 1.0)	3	8280.940
double__well	LSC-CP	c1.0__10.1 (scale 1.0)	3	6430.840
double__well	LSC-CP	c1.0__10.3 (scale 1.0)	3	6971.247
double__well	LSC-CP	c1.0__11.0 (scale 1.0)	3	6693.898
double__well	LSC-CP	c1.0__13.0 (scale 1.0)	3	9923.369
double__well	LSC-CP	c1.25__10.05 (scale 1.25)	3	9164.400
double__well	LSC-CP	c1.25__10.1 (scale 1.25)	3	6323.230
double__well	LSC-CP	c1.25__10.3 (scale 1.25)	3	7051.971
double__well	LSC-CP	c1.25__11.0 (scale 1.25)	3	6745.983
double__well	LSC-CP	c1.25__13.0 (scale 1.25)	3	11464.567
double__well	LSC-CP	c1.5__10.05 (scale 1.5)	3	9609.052
double__well	LSC-CP	c1.5__10.1 (scale 1.5)	3	6442.918
double__well	LSC-CP	c1.5__10.3 (scale 1.5)	3	7358.622
double__well	LSC-CP	c1.5__11.0 (scale 1.5)	3	7094.470
double__well	LSC-CP	c1.5__13.0 (scale 1.5)	3	12495.297

Table 12: Generated metrics for double well scale intensity relax. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double__well	LSC-CP	c0.5__10.05 (scale 0.5)	3	2594.977
double__well	LSC-CP	c0.5__10.1 (scale 0.5)	3	2708.140
double__well	LSC-CP	c0.5__10.3 (scale 0.5)	3	2687.791
double__well	LSC-CP	c0.5__11.0 (scale 0.5)	3	2664.856
double__well	LSC-CP	c0.5__13.0 (scale 0.5)	3	5014.708
double__well	LSC-CP	c0.75__10.05 (scale 0.75)	3	2666.340
double__well	LSC-CP	c0.75__10.1 (scale 0.75)	3	2680.166
double__well	LSC-CP	c0.75__10.3 (scale 0.75)	3	2691.575
double__well	LSC-CP	c0.75__11.0 (scale 0.75)	3	2826.113
double__well	LSC-CP	c0.75__13.0 (scale 0.75)	3	6613.779
double__well	LSC-CP	c1.0__10.05 (scale 1.0)	3	2708.096
double__well	LSC-CP	c1.0__10.1 (scale 1.0)	3	2730.528
double__well	LSC-CP	c1.0__10.3 (scale 1.0)	3	2733.476
double__well	LSC-CP	c1.0__11.0 (scale 1.0)	3	3311.263
double__well	LSC-CP	c1.0__13.0 (scale 1.0)	3	9469.193
double__well	LSC-CP	c1.25__10.05 (scale 1.25)	3	2729.629
double__well	LSC-CP	c1.25__10.1 (scale 1.25)	3	2752.456
double__well	LSC-CP	c1.25__10.3 (scale 1.25)	3	2752.854
double__well	LSC-CP	c1.25__11.0 (scale 1.25)	3	3502.172
double__well	LSC-CP	c1.25__13.0 (scale 1.25)	3	11122.779
double__well	LSC-CP	c1.5__10.05 (scale 1.5)	3	2740.748
double__well	LSC-CP	c1.5__10.1 (scale 1.5)	3	2751.411
double__well	LSC-CP	c1.5__10.3 (scale 1.5)	3	2775.734
double__well	LSC-CP	c1.5__11.0 (scale 1.5)	3	3599.659
double__well	LSC-CP	c1.5__13.0 (scale 1.5)	3	12078.285

Table 13: Generated metrics for double well timestep dt000075. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	2988.754
double_well	raw CP	double_well_shell (scale 1.0)	4	7382.971
double_well	local	none	4	31889.311

Table 14: Generated metrics for double well timestep dt00015. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	5162.676
double_well	raw CP	double_well_shell (scale 1.0)	4	15023.003
double_well	local	none	4	49290.767

Table 15: Generated metrics for double well timestep dt0003. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	17052.948
double_well	raw CP	double_well_shell (scale 1.0)	4	47754.173
double_well	local	none	4	88340.757

Table 16: Generated metrics for double well timestep dt0006. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	ESS/sec
double_well	LSC-CP	double_well_shell (scale 1.0)	4	45772.913
double_well	raw CP	double_well_shell (scale 1.0)	4	88278.867
double_well	local	none	4	2.61e+05

Table 17: Generated metrics for four well graph. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin KL	mode/basin-TV	ESS/sec	ESS/grad
four_well	BAOAB	none	3	1.122	0.585	549.506	0.097
four_well	HMC	none	3	0.028	0.102	114.296	0.017
four_well	LSC-CP	minima_complete_graph (scale 1.0)	3	0.002	0.027	296.603	0.433
four_well	LSC-CP	minima_edge_graph (scale 1.0)	3	0.003	0.032	220.053	0.318
four_well	LSC-CP	random_matched_length_control	3	0.018	0.081	511.978	0.740
four_well	MALA	none	3	0.275	0.312	182.343	0.049
four_well	PT	none	3	0.004	0.035	3201.332	0.264
four_well	raw CP	minima_complete_graph (scale 1.0)	3	0.002	0.026	776.539	0.472
four_well	raw CP	minima_edge_graph (scale 1.0)	3	0.003	0.030	538.809	0.327
four_well	raw CP	random_matched_length_control	3	0.185	0.257	1179.940	0.716
four_well	local	none	3	0.275	0.305	532.329	0.099

Table 18: Generated metrics for manywell scaling. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	block KL	count KL	deep-count mean	ESS/sec	Levy evals
manywell_d16	BAOAB	none	3	3.605	4.874	0.075	1335.609	0.000
manywell_d16	HMC	none	3	2.938	4.874	0.156	266.894	0.000
manywell_d16	LSC-CP	manywell_block_flip	3	0.002	0.015	6.865	562.973	2.95e+08
manywell_d16	MALA	none	3	3.608	4.874	0.076	424.720	0.000
manywell_d16	PT	none	3	0.431	3.269	3.161	331.966	0.000
manywell_d16	raw CP	manywell_block_flip	3	0.254	2.067	4.044	624.517	0.000
manywell_d16	local	none	3	1.448	4.810	0.891	1272.990	0.000
manywell_d32	BAOAB	none	3	3.585	4.509	0.153	1358.250	0.000
manywell_d32	HMC	none	3	2.910	4.509	0.324	269.894	0.000
manywell_d32	LSC-CP	manywell_block_flip	3	0.003	0.056	13.885	476.763	5.90e+08
manywell_d32	MALA	none	3	3.761	4.509	0.126	413.177	0.000
manywell_d32	PT	none	3	0.505	4.112	5.733	335.205	0.000
manywell_d32	raw CP	manywell_block_flip	3	0.255	3.075	8.076	649.638	0.000
manywell_d32	local	none	3	1.232	4.509	2.303	1290.182	0.000
manywell_d64	BAOAB	none	3	3.822	4.175	0.309	1534.993	0.000
manywell_d64	HMC	none	3	2.932	4.175	0.633	294.039	0.000
manywell_d64	LSC-CP	manywell_block_flip	3	0.008	0.238	28.286	447.920	1.18e+09
manywell_d64	MALA	none	3	4.030	4.175	0.251	447.540	0.000
manywell_d64	PT	none	3	0.573	4.141	10.488	358.454	0.000
manywell_d64	raw CP	manywell_block_flip	3	0.242	3.877	16.462	540.720	0.000
manywell_d64	local	none	3	0.925	4.175	6.714	1413.954	0.000
manywell_d8	BAOAB	none	3	4.508	5.198	0.043	1467.754	0.000
manywell_d8	HMC	none	3	2.867	5.128	0.083	281.569	0.000
manywell_d8	LSC-CP	manywell_block_flip	3	8.58e-04	0.004	3.399	614.640	1.47e+08
manywell_d8	MALA	none	3	3.915	5.179	0.029	468.775	0.000
manywell_d8	PT	none	3	0.369	1.551	1.719	342.175	0.000
manywell_d8	raw CP	manywell_block_flip	3	0.258	1.008	2.010	710.733	0.000
manywell_d8	local	none	3	1.616	4.601	0.363	1323.734	0.000

Table 19: Generated metrics for muller10d basin communication. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV	coverage time	coverage frac	ESS/sec
muller10d	LSC-CP	minima_complete_graph (scale 1.0)	4	0.012	1.911	0.636	5985.218
muller10d	LSC-CP	random_matched_length_control	4	0.124	2.367	0.152	1774.847
muller10d	raw CP	minima_complete_graph (scale 1.0)	4	0.181	1.818	0.697	4645.951
muller10d	raw CP	random_matched_length_control	4	0.030	2.274	0.296	3490.095
muller10d	local	none	4	0.110	2.410	0.155	5901.819

Table 20: Generated metrics for muller10d benchmark. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV	ESS/sec	ESS/grad
muller10d	BAOAB	none	4	0.219	7245.638	5.266
muller10d	HMC	none	4	0.067	1353.333	0.654
muller10d	LSC-CP	minima_complete_graph (scale 1.0)	4	0.012	5937.556	13.997
muller10d	MALA	none	4	0.156	2598.186	2.133
muller10d	PT	none	4	0.100	2546.191	0.592
muller10d	raw CP	minima_complete_graph (scale 1.0)	4	0.181	4485.277	5.254
muller10d	local	none	4	0.110	5635.440	4.238

Table 21: Generated metrics for muller10d invariance. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV
muller10d	LSC-CP	minima_complete_graph (scale 1.0)	4	0.012
muller10d	raw CP	minima_complete_graph (scale 1.0)	4	0.191
muller10d	local	none	4	0.019

Table 22: Generated metrics for triple well benchmark. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	mode/basin-TV	ESS/sec	ESS/grad
triple_well	BAOAB	none	4	0.266	1631.041	0.490
triple_well	HMC	none	4	0.009	685.008	0.223
triple_well	LSC-CP	adjacent (scale 1.0)	4	0.047	278.960	0.521
triple_well	MALA	none	4	0.007	1548.820	0.771
triple_well	PT	none	4	0.018	68096.255	9.246
triple_well	raw CP	adjacent (scale 1.0)	4	0.276	613.699	0.503
triple_well	local	none	4	0.739	2687.622	0.814

Table 23: Generated metrics for triple well invariance. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	basin-TV	CDF-sup
triple_well	LSC-CP	adjacent (scale 1.0)	4	0.011	0.019
triple_well	raw CP	adjacent (scale 1.0)	4	0.165	0.188
triple_well	local	none	4	0.009	0.016

Table 24: Generated metrics for triple well support. Means and standard errors are computed from raw CSV rows.

Target	Method	Bank	n	mode/basin-TV	middle-mass err	coverage frac	ESS/sec
triple_well	LSC-CP	adjacent (scale 1.0)	4	0.047	0.011	0.573	282.853
triple_well	LSC-CP	overlong (scale 1.0)	4	0.015	0.017	0.864	707.074
triple_well	LSC-CP	short (scale 0.4)	4	0.375	0.075	0.074	268.894
triple_well	raw CP	adjacent (scale 1.0)	4	0.276	0.170	0.368	706.769
triple_well	raw CP	overlong (scale 1.0)	4	0.168	0.163	0.445	742.813
triple_well	raw CP	short (scale 0.4)	4	0.454	0.151	0.068	699.504
triple_well	local	none	4	0.739	0.394	0.000	2784.303